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MILITARY SPECIFICATION

COMPRESSORS, HERMETIC,

FOR

AIR CONDITIONER APPLICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hermetic compressors used in military designed air conditioners.

1.2 Classification. The compressors shall be of the following types, classes and styles, as specified (see 6.2):

Type I - 6,000 Btu/Hr nominal capacity
Class I - 115 Volt, 50/60 Hertz-single-phase
Class II - 208 Volt, 400 Hertz-three-phase
Style I - Refrigerant 12
Style II - Refrigerant 22

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of thid document or by letter.

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Type II - 9,000 Btu/Hr nominal capacity
Class I - 115 Volt, 50/60 Hertz-single-phase
Class II - 230 Volt, 50/60 Hertz-single-phase
Class III - 208 Volt, 50/60 Hertz-three-phase
Class IV 208 Volt, 400 Hertz-three-phase
Style II Refrigerant 22

Type III - 18,000 Btu/Hr nominal capacity
Class I 230 Volt, 50/60 Hertz-single-phase
Class II 208 Volt, 50/60 Hertz-three-phase
Class III 208 Volt, 400 Hertz-three-phase
Style II Refrigerant 22

Type IV - 36,000 Btu/Hr nominal capacity
Class I 208 Volt, 50/60 Hertz-three-phase
Class II 208 Volt, 400 Hertz-three-phase
Style II Refrigerant 22

Type V - 60,000 Btu/Hr nominal capacity
Class I 208 Volt, 50/60 Hertz-three-phase
Class II 208 Volt, 400 Hertz-three-phase
Style I Refrigerant 12
Style II Refrigerant 22

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

O-F-499	- Flux, Brazing, Silver Alloy, Low Melting Point.
BB-F-1421	- Fluorocarbon Refrigerants.
QQ-B-654	- Brazing Alloys, Silver.
QQ-S-781	- Strapping, Steel and Seals.
VV-L-825	- Lubricating Oil, Refrigerant Compressor.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

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MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-W-5086	- Wire, Electric, Polyvinyl Chloride Insulated, Copper or Copper Alloy.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum, and Aluminum Alloys.
MIL-A-52424	- Air Conditioners: Wall - Base-Mounted, Self-Contained, Multipackage, Standard Weight, Air Cooled, 6,000 Btu/Hr.
MIL-A-52435	- Air Conditioner: Base Mounted, Air-Cooled, 208-Volt, 3-Phase, 50- and 60-Hertz, AC, Single-Package, 36,000 Btu/Hr.
MIL-A-52767	- Air Conditioners: Vertical and Horizontal, Compact.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment, and Facilities.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AIR CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

ARI Standard 515 -Sealed Refrigerant Compressors and Condensing Units, Twenty Horsepower and Smaller.

(Application for copies should be addressed to the Air Conditioning and Refrigeration Institute, 1815 North Fort Myer Drive, Arlington, VA 22209.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standard MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE STD 43 - Testing Insulation Resistance of Rotating Machinery.

Bulletin "Communications and Electronics", 1955 by R. E. Seeley.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS (ASHRAE)

Standard 23 - Methods of Testing for Rating Refrigerant Compressors.

(Application for copies should be addressed to the American Society of Heating, Refrigerating, and Air Conditioning Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The compressors shall be of a hermetically sealed type as defined by ARI Standard 515, designed for air conditioning systems which employ Refrigerant 22 (monochlorodifluoromethane - $CHClF_2$) or Refrigerant 12 (dichlorodifluoromethane - CCl_2F_2). The compressor shall be used in military air conditioners specified in table I.

3.2 Qualification. Compressors furnished under this specification shall be products which have been tested, have passed the qualification tests specified herein, and have been listed on or approved for listing on the applicable qualified products list (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The compressor shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the compressor may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the compressor may be newly fabricated from recovered materials to the maximum extent practicable, provided the compressor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the compressor.

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3.4 Performance characteristics.

3.4.1 Capacity. When operating on 60 hertz or 400 hertz electrical power as applicable and at the conditions specified in 4.5.2.2, the capacity of the compressor shall be as shown in table I, and when installed in the air conditioner(s) shall meet the capacity requirements of the intended air conditioner (see table V).

3.4.1.1 Capacity on 50 hertz power. When operating on 50 Hertz power and rated voltage, all 50/60 hertz compressors shall produce at least 80 percent of the minimum capacity specified in table I.

TABLE I. Compressor capacity and power requirement.

Compressor	Maximum / Minimum Capacity Btu/hr	Minimum Btu/Watt	Maximum Power kW
1	2	3	4
Type I Class I, Style I	6,500 5,900	4.8	1.4
Class I, Style II	8,600 7,400	5.3	1.6
Class II, Style I	6,700 6,200	4.8	1.5
Class II, Style II	8,600 7,400	5.3	1.6
Type II Class I, Style II	13,100 11,000	5.7	2.3
Class II, Style II	13,100 11,000	6.2	2.1
Class III, Style II	13,100 11,000	5.7	2.3
Class IV, Style II	13,100 11,000	5.9	2.2
Type III Class I, Style II	26,000 22,200	6.1	4.2
Class II, Style II	26,000 22,200	6.1	4.2
Class III, Style II	26,000 22,200	5.9	4.4
Type IV Class I, Style II	49,000 41,500	6.5	7.5
Class II, Style II	49,000 41,500	6.1	8.0

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TABLE I. Compressor capacity and power requirement. (Cont'd)

Compressor	Maximum / Minimum Capacity Btu/hr	Minimum Btu/Watt	Maximum Power kW
1	2	3	4
Type V Class I, Style II	76,000 62,000	6.3	12.0
Class II, Style II	76,000 62,000	5.8	13.0

3.4.2 Power consumption. When the compressor is operating at the conditions specified in 3.4.1, the power consumption shall be not greater than as specified in table I. The performance factor Btu/Watt shall be not less than as specified in table I. The compressors shall be designed to utilize the power factor correcting capacitors specified in the Technical Data Packages of the specifications listed in table V.

3.4.3 Operating temperatures. The compressors shall be capable of operation in the intended air conditioner at the conditions specified for high temperature operation in the applicable air conditioner specifications, MIL-A-52424, MIL-A-52435 and MIL-A-52767, and shall be capable of operation at the conditions specified in table II. The air velocity across the compressor shall be 500 fpm (feet per minute) ±100 fpm.

TABLE II. Compressor design operating temperature conditions.

Saturated Discharge Temp. ±1 °F	Saturated Suction Temp. °F	Compressor Ambient Temp. ±2 °F
160	40 to 60	160
150	35 to 60	150
140	30 to 60	140
130	25 to 55	130
120	20 to 50	120
100	15 to 40	100
60	(-) 20 to 10	60

3.4.4 Starting characteristics. The compressors shall start and operate when electrically connected to the circuit breaker on the intended military air conditioner (see table V), without tripping the circuit breaker or damage to the compressor.

3.4.5 Reliability. The Mean-Time-Between-Failures (MTBF) shall be as follows when tested as specified in 4.5.2.5:

Upper Test MTBF (0_o) = 2430 hours
Lower Test MTBF (0_1) = 1215 hours

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3.4.6 Tilted operation. The compressor shall start and operate without malfunction when the base of the compressors are positioned 10 degrees from the horizontal in any plane.

3.4.7 Refrigerant leakage. When the compressors are subjected to the test specified in 4.5.2.7 the refrigerant leakage shall be not more than 0.1 ounce per year.

3.4.8 Rupture pressure. Style I compressors shall withstand pressures up to 250 pounds per square inch gauge (psig), and style II compressors shall withstand pressures of 470 psig without permanent deformation. The high side section of style I compressors shall withstand pressures of 1175 psig, and low side section of style I shall withstand 350 psig pounds per square inch without rupture. The high side section of style II compressors shall withstand pressures of 1500 psig and the low side section of style II shall withstand 612.5 psig without rupture.

3.4.9 Vibration. When subjected to the test specified in 4.5.2.8 the compressor shall show no evidence of damage, failure, or malfunction.

3.4.10 Electromagnetic interference. The electromagnetic interference emission and susceptability characteristics of the compressor when installed in the intended air conditioner, shall conform to MIL-STD-461 when tested as specified in 4.5.2.9.

3.4.11 Salt-fog, and humidity. The compressor shall show no peeling of paint, corrosion, refrigerant leaks, damage, electrical shorts, when subjected to salt-fog, and humidity tests specified in 4.5.2.11 and 4.5.2.12.

3.4.12 Storage temperature. The compressors shall not be damaged and their operation shall not be impaired by storage at an ambient temperature of +155 °F for 4 hours, no solar radiation, and at -65 °F for 12 hours.

3.4.13 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish in accordance with the requirements of MIL-V-173, composition I, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements, whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

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3.4.14 Air conditioner compatability (weight, dimensions, connection sizes and mounting). Compressors shall be physically and electrically compatible with the applicable military air conditioner (see table V) for which the compressor is designed without requiring any changes or modifications to the components, mounting, electrical circuit, or structure of the air conditioner, except for piping to common interface points.

3.4.15 Suction line strainer. A suction line strainer (40 mesh or finer screen) may be provided in or on all compressor assemblies.

3.4.16 Crankcase heater. Compressors shall be provided with a crankcase heater to assure proper operation of the compressors at the operating temperatures specified in 3.4.3.

3.4.17 Wiring insulation. All wiring within the compressor junction box shall be insulated in accordance with MIL-W-5086 for class I or class II insulation.

3.4.18 High potential. The compressor motor shall be capable of withstanding, for 1 minute without breakdown, the application of 60 hertz (Hz) potential between high-voltage parts and dead-metal parts, and between parts of high and low voltage circuits. The potential shall be:

a. 1,000 volts plus twice rated voltage, except as noted below.

b. 900 volts for motors rated not more than 250 volts and having locked-rotor kilovolt-ampere values not exceeding 6.76 for single phase motors and 4.76 for three phase motors.

3.4.19 Motor protection. Compressor assemblies shall be provided with temperature protection to prevent damage to the motor or compressor resulting from operating temperature conditions higher than recommended by the manufacturer. The manufacturer shall certify and provide maximum average motor winding temperatures as determined by Seeley Circuit Tester and maximum discharge gas temperatures.

3.4.20 Voltage frequency variation. The compressor shall be capable of startup and operation when tested as specified in 4.5.2.15.

3.5 Lubricants. The compressor shall be filled to the operating level with oil conforming to VV-L-825, or as recommended by the manufacturer.

3.6 Identification marking. The compressor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the compressor on plates conforming to MIL-P-514, type I, style 2, composition C, of type I, grade A, class 1 material. Plates shall be attached in a conspicuous protected location, and shall contain the following:

Type Refrigerant
Compressor Manufacturer
Compressor Serial No.
Compressor Manufacturer's Model No.

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Volts, Frequency/Phase
Type Oil
Oil Charge
NSN
Full Load Current
Lock Rotor Current

3.7 Wiring diagram. A wiring diagram which incorporates all electrical circuits from the power source to the motor shall be furnished on each compressor.

3.8 Human factors engineering. The compressor shall conform to Human Factors Engineering Criteria as described in MIL-STD-1472, 5.5 (Labeling) and 5.13 (Hazards and Safety), as applicable.

3.9 Treatment and painting. The portions of the compressors normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.10 Workmanship.

3.10.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.10.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.10.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.10.4 Cleanliness. The internal parts of the compressor shall be free from scale, paint, grease, oxide film, moisture, and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 Qualification inspection. A qualification inspection shall be performed as facilities acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.3.1 Sample. A sample of four compressors shall be examined as specified in 4.3.2. The sample shall be taken at random from a production run and shall be produced with equipment and procedures normally used in production. The sample units shall have been subjected to and passed the requirements of the inspection (see 4.3.2). Qualification shall not be granted if examination requirements are not met. Each compressor shall be qualified separately.

4.3.2 Inspection routine. Sample units shall be subjected to the inspection tests marked "X" in the qualification inspection column of table III, in the order shown. Sample size shall be as specified in 4.3.1.

TABLE III. Inspection Schedule.

Inspection Sequence	Type of Inspection: Qualification Inspection				Qual. Conf.	Examination or Test	Requirement	Test
	Unit 1	Unit 2	Unit 3	Unit 4				
1	2	3	4	5	6	7	8	9
1	X	X	X	X	X	Examination	3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9	4.5.1
2	X	X	X	X	X	Leak test	3.4.7	4.5.2.7.1

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TABLE III. Inspection Schedule. (Cont'd)

Inspection Sequence	Type of Inspection: Qualification Inspection Unit 1	Unit 2	Unit 3	Unit 4	Qual. Conf.	Examination or Test	Requirement	Test
1	2	3	4	5	6	7	8	9
3	X	X	X	X	X 1/	Thermal protector	3.4.19	4.5.2.14
4	X	X	X	X	X 1/	Compressor calorimeter	3.4.1, 3.4.2	4.5.2.2
5	-	-	X	-	-	Capacity	3.4.1	4.5.2.3
6	-	-	X	-	-	High temperature operation	3.4.3	4.5.2.4
7	-	-	X	-		Tilted operation	3.4.6	4.5.2.6
8	-	-	X	-		Voltage and frequency variation	3.4.20	4.5.2.15
9	-	X	-	-	X	High potential	3.4.18	4.5.2.13
10	-	-	-	X	-	Electromagnetic	3.4.10	4.5.2.9
11	X 2/	-	-	-	-	High and Low temp strorage	3.4.12	4.5.2.10
12	X 2/	-	-	-	-	Humidity	3.4.11	4.5.2.12
13	X 2/	-	-	-	-	Salt fog	3.4.11	4.5.2.11
14	X	-	-	-	-	Vibration	3.4.9	4.5.2.8
15	X	-	-	-	-	Leak test	3.4.7	4.5.2.7.1
16	X	X	X	X	-	Reliability	3.4.5	4.5.2.5
17	X	X	X	X	-	Compressor Calorimeter	3.4.1	4.5.2.2
18	X	-	X	-	X 1/	Rupture	3.4.8	4.5.2.7.2
19	-	-	-	-	X	Human Factors	3.8	4.5.1

1/ Test conducted on first item produced for each contract or order consisting of a minimum quantity of 20 units.
2/ Test may be conducted on unit 2 in lieu of unit 1.

4.3.3 Failures. Failures of any test or examination shall be cause for rejection.

4.3.4 Inspection results report. A summary of the inspection results of the tests performed shall be furnished to the qualifying activity for approval (see 6.3).

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4.3.5 Acceptance criteria. Acceptance by the Government shall be based not only on the compressor during the entire series of tests and examinations but also on the condition of the compressor and its components which shall be examined by teardown at the completion of the tests. The following components shall be disassembled and examined in detail:

a. Internal mounting components.
b. Suction and discharge valves.
c. Cylinders.
d. Bearings.
e. Internal tube fittings.
f. Electrical connections and components.
g. Piston.
h. Seals.
i. Shaft.

Additional examination shall be made at the option of the Government. Each evidence of permanent deformation, breakage, excessive wear of any part or component shall be cause for rejection.

4.3.6 Retention of qualification. To retain qualification, the contractor shall requalify the compressor at 5-year intervals. Requalification shall be as specified in 4.3 through 4.3.5.

4.4 Quality conformance inspection.

4.4.1 Examination. Each compressor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each compressor shall be subjected to the tests marked "X" in column 5 of table III except as noted. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The compressor shall be examined for the following defects:

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatments.
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the compressor.
106. Physical dimensions not compatible with specified air conditioner.
107. Weight of compressor (charged with oil) not as specified.
108. Compressor not charged with oil.
109. Suction-line strainer not as specified, if provided.
110. Lubricants not as specified.

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111. Electrical connector not as specified in applicable air conditioner drawings.
112. Identification markings not as specified.
113. Wiring diagram not as specified.
114. Compressor not hermetic type.
115. Electrical circuitry not protected from the effects of moisture- and fungus-growth as specified.
116. Bolted and riveted connections not as specified.
117. Metal fabrication not as specified.
118. Treatment and painting not as specified.
119. Cleanliness of compressors internal parts not as specified.
120. Wiring insulation not as specified.
121. Paint color not as specified.
122. Human factors engineering not as specified.
123. Crankcase heater not provided.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted employing the electric power for which the compressors are capacity rated.

4.5.2.2 Compressor calorimeter capacity. Tests shall be conducted on a compressor calorimeter utilizing the methods and procedures outlined in ASHRAE 23. The capacity rating conditions shall be as follows:

Test conditions: Saturated discharge temperature - 150 ± 1.0 °F.
Subcooling - 10 °F minimum
Saturated suction temperature - 50 ± 0.5 °F.
Superheat - 25 +5 or -0 °F
Ambient air over compressor - 150 ±5 °F.
Rated voltage.

Power factors shall be sufficiently high to comply with the requirements of the specifications listed in table V.

Nonconformance to 3.4.1 and 3.4.2 shall constitute failure of this test.

4.5.2.2.1 Compressor calorimeter capacity (50 hertz power). When 50/60 hertz compressors are specified, test the compressor as required in 4.5.2.2 except that 50 hertz power shall be utilized. Nonconformance with the 50 hertz requirement of 3.4.1.1 shall constitute failure of this test.

4.5.2.3 Compressor - air conditioner capacity. Install the compressor in the applicable air conditioner. Test the air conditioner for cooling capacity at the capacity rating conditions and procedures of the applicable air conditioner specification. Nonconformance to the performance requirements of the air conditioner specification shall constitute failure of this test.

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4.5.2.4 High temperature operation. Install the compressor in the air conditioner and operate on cool mode at the conditions specified for high temperature operation in the applicable specification of the intended air conditioner, see table V. The following data shall be recorded:

a. Volts.
b. Amperes.
c. Refrigerant pressures (suction and discharge).
d. Refrigerant temperatures:
    (1) Compressor discharge.
    (2) Liquid line.
    (3) Suction line.
e. Air temperatures:
    (1) Entering and leaving evaporator (DB).
    (2) Entering condenser (DB).
f. Crankcase heater wattage.

Nonconformance to the high temperature and low temperature performance requirements of the air conditioner specification shall constitute failure of this test.

4.5.2.5 Reliability. Four of the qualification inspection compressors shall be tested in a compressor loop endurance facility at the conditions of table IV for a total of 6640 hours. Failed compressors will be replaced and testing continued until the total test time is reached. Four or more failures as defined in 6.5 shall constitute failure of the reliability test.

4.5.2.5.1 Test facilities and apparatus. The test facilities and apparatus used in conducting the tests shall be as specified and capable of maintaining the conditions required in table IV when testing hermetic compressors from 6,000 Btuh to 60,000 Btuh nominal capacity. Means shall be provided to equalize pressure from low to high side. Equalization time shall not exceed 3 minutes. The test apparatus shall be capable of maintaining not less than 500 ft/min air velocity across the compressor at the ambient temperature specified in table IV. The system design shall return oil at the lowest load condition and shall include a suction line accumulator, properly sized to prevent oil trapping, to minimize the possibility of liquid slugging. A sight glass shall be installed in the suction line downstream of the suction line accumulator and within 24 inches of the compressor suction connection for observation of possible slugging. The test apparatus shall contain a filter/drier in the liquid line, and a suction line filter located in the suction line between the sight glass and compressor inlet connection. Both filter elements shall be replaced whenever the system is opened to atmosphere. The facilities shall be capable of testing both low side and high side cooled compressors. The high side cooled compressor facilities shall be capable of maintaining the precooler temperature indicated in table IV with either a heat exchanger or by means of sufficient air velocity across the precooler furnished as part of the compressor. The system shall contain the following protective devices containing provisions to indicate which device (including compressor motor protector) was activated during abnormal operation:

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a. High pressure cut-out.
b. Low pressure cut-out.
c. Discharge temperature cut-out.
d. Discharge pressure relief valve.
e. Over-current protector.

4.5.2.5.1.1 Instrumentation. The accuracy of instruments and test equipment used to control or monitor the test parameters shall be verified every 6 months. All instruments and test equipment shall have the accuracy and tolerance specified herein. (Test instruments shall be of the continuous recording type.) All instruments used for electrical measurements will have an accuracy equal to or better than 2 percent of the reading. Refrigerant pressure gauges shall have an error of not more than 1 percent of full scale. The smallest division of the high pressure gauge shall not exceed 2 psig and the low side shall not exceed 1 psig. Refrigerant temperatures shall be measured with thermocouples. The smallest scale division of the potentiometer shall not exceed 5 °F. The following data shall be recorded during performance of the test:

a. Suction pressure within 24 inches of compressor.
b. Discharge pressure within 24 inches of compressor.
c. Suction temperature within 12 inches of compressor.
d. Discharge temperature within 12 inches of compressor.
e. Amperage.
f. Ambient temperature.
g. Voltage.
h. Wattage.

4.5.2.5.1.2 Test performance. The tests shall be conducted employing the electric power for which the compressor is rated and as specified in table IV. No braces, sealing materials, tiedown devices, or other items that are not inherently a part of the operating compressor shall be employed. No alignment of components or other servicing shall be performed on the compressor after testing has been initiated. Prior to test the compressor shall be examined for proper oil type and quantity. The test shall be performed at the condition specified in table IV. Refrigerant used shall conform to BB-F-1421. After not less than 1 hour of continuous operation, withdraw a sample of refrigerant from the system and determine the moisture content by one of the following:

a. Methods specified in BB-F-1421.
b. Electrical method employing hygroscopic film.

The system shall contain not more than 25 ppm (parts per million) by weight of moisture.

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TABLE IV. Compressor loop endurance test schedule.

Test Run	Min. Operating Hrs.	Min. Test Hrs.	Volts	Hz	Saturated Discharged Temp °F	Saturated Suction Temp °F	Suction Temp. °F	Comp. Ambient °F	Cycling
1*	132	168	Rated	Rated	150	50	75	150	15 min on, 5 off
2*	132	168	Rated	Rated	110	30	55	110	15 min on, 5 off
3*	132	168	Rated	Rated	125	35	60	125	15 min on, 5 off
4	336	336	Rated	Rated	125	35	60	125	-
5	336	336	Rated	Rated	110	30	55	110	-
6	300	300	Rated	Rated	160	60	85	160	-
7	48	48	Rated	Rated	150	50	75	150	-
8*	132	168	Rated	Rated	150	50	75	150	15 min on, 5 off
9	48	48	+10%	Rated	150	50	75	150	-
10	48	48	-10%	Rated	150	50	75	150	-
11	8	8	-5%	50 Hz	150	50	75	150	-
12	8	8	+5%	50 Hz	150	50	75	150	-
Total	1660	1804							

*No cycling for first 24 hours of operation.

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4.5.2.6 Tilted operation. Operate the compressor in the air conditioner for a period of not less than 6 hours in an ambient temperature of +80 ±3 °F dry coil condition dry bulb with air at the same temperature entering the evaporator. During this test, operate the air conditioner with the bottom at a slope of 10 degrees from the horizontal in any plane. For each 1-1/2 hours of operation, the bottom edge of the four sides shall be raised, in turn, to form the 10-degree slope. Evidence of malfunction, rough operation, and vibration shall constitute failure of this test.

4.5.2.7 Refrigerant leakage and rupture.

4.5.2.7.1 Refrigerant leakage. Charge the compressor with refrigerant gas detectable with a halogen sensitive electronic leak detector (General Electric Type H-2 or equal) to a pressure of not less than 120 psig. Calibrate the leak detector to indicate a leakage rate of 0.1 ounce per year. Check the entire compressor for leaks. Pressure shall be maintained during the leak search. Any detectable refrigerant leak shall constitute failure of this test.

4.5.2.7.2 Rupture pressure. Test pressure tolerances shall be +5 and -0 psig. Purge air from the compressor by filling the compressor with oil and connect the system to a hydraulic pump. Raise the pressure gradually simultaneously on the high and low pressure sides until the following respective pressure is reached. Maintain the pressure for 1 minute.

a.	Style I high side section	1175 psig
b.	Style I low side section	350 psig
c.	Style II high side section	1500 psig
d.	Style II low side section	612.5 psig

Rupture of the compressor shall constitute failure of this test.

4.5.2.8 Vibration. Conduct the vibration test as specified in MIL-STD-810, method 514.2, category F, procedure VIII, table 514.2-VI, figure 514.2-6, curve V. The test time shall be 60 minutes total per plane. The compressor shall not be operated during this test. The compressor shall be rigidly mounted on a support plate which shall be mounted on the vibration machine. During the tests the base of the compressor shall remain in a horizontal plane. After completion of all vibration tests, conduct the calorimeter test specified in 4.5.2.2. Any evidence of internal or external damage or malfunction shall constitute failure of this test.

4.5.2.9 Electromagnetic interference. The compressor shall be tested in the intended air conditioner (see table V) in accordance with the test requirements of the applicable air conditioner to determine conformance to 3.4.10.

4.5.2.10 High- and low-temperature storage. Conduct insulation resistance test specified in IEEE STD 43.

4.5.2.10.1 Low-temperature. Place the compressor in an ambient temperature of -65 +0 or -2 °F for a continuous period of 12 hours. At the conclusion of

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the exposure period, remove the compressor from the test chamber and allow to soak at room ambient temperature for a period of not less than 1 hour. Conduct refrigerant leakage test specified in 4.5.2.7.1 and repeat insulation resistance test.

4.5.2.10.2 High-temperature. Subject the compressor to a temperature of +155 °F for a period of 4 hours. At the conclusion of the exposure period, remove the compressor from the test chamber and examine for deterioration, corrosion, or damage. After completion of the examination, allow the compressor to soak at room ambient temperature for a period of not less than 1 hour but not more than 3 hours. After completion of the soaking period, repeat the leakage and insulation resistance tests.

4.5.2.10.3 Failure criteria. The following shall constitute failure of this test:

a. Any evidence of deterioration or damage.
b. Failure of leakage test.
c. Variation in insulation resistance.

4.5.2.11 Salt fog. Pressurize compressor with nitrogen to a pressure of 5 to 10 psig. Conduct insulation resistance test specified in IEEE STD 43. Subject the compressor to test method 509, procedure I of MIL-STD-810. Do not operate the compressor during exposure. Upon completion of the test conduct refrigerant leakage test specified in 4.5.2.7.1 and repeat insulation resistance test. The following shall constitute failure of this test:

a. Any evidence of deterioration or damage.
b. Failure of leakage test.
c. Variation in insulation resistance.

4.5.2.12 Humidity. Pressurize compressor with nitrogen to a pressure of 5 to 10 psig. Conduct insulation resistance test specified in IEEE STD 43. Subject the compressor to test method 507, procedure I of MIL-STD-810. Do not operate the compressor during exposure. Upon completion of the test conduct refrigerant leakage tests specified in 4.5.2.7.1 and repeat insulation resistance test. The following shall constitute failure of this test:

a. Any evidence of deterioration or damage.
b. Failure of leakage test.
c. Variation in insulation resistance.

4.5.2.13 High potential test. The compressor assembly shall be tested in accordance with NEMA MG1-12-02. Nonconformance to 3.4.18 shall constitute failure of this test.

4.5.2.14 Thermal protector.

4.5.2.14.1 Apparatus. This test may be performed with the compressor operating on either a compressor calorimeter or an endurance loop.

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4.5.2.14.2 Data. Data to be recorded shall include but not be limited to the following:

a. Discharge pressure.
b. Suction pressure.
c. Suction gas temperature entering compressor.
d. Compressor ambient air temperature.
e. Discharge gas temperature leaving compressor.
f. Volts.
g. Amps.
h. Motor winding resistance (see 4.5.2.14.3, step a.)

4.5.2.14.3 Procedure. The procedure shall be as follows:

a. Measure the compressor motor winding resistance and temperature with a Seeley Circuit Tester as described in IEEE Bulletin "Communications and Electronics", 1955.
b. Install the compressor on the test apparatus and connect the Thermal Protection Device as follows:

(1) Integral Type Protector - This type of protector senses both temperature and line current to the compressor motor and disconnects the power supply to the motor upon actuation. It is normally located inside the compressor terminal box. For this test this protector must be wired electrically into the compressor motor circuit before the "Seeley Circuit Tester".
(2) Pilot Type Protector - This type protector normally senses temperature only and does not directly disconnect the power supply to compressor motor upon actuation. It is normally located internally on the motor windings. For this test this type protector shall be wired to activate an external alarm system when the protector actuates.

c. Operate the compressor at the following initial conditions:

Saturated discharge temperature	150 $\pm$ 0.5 °F
Saturated suction temperature	50 $\pm$ 1.0 °F
Superheat at compressor	25 $\pm$ 2.0 °F
Ambient air over compressor	150 $\pm$ 2.0 °F

Continue operation at these conditions for a minimum of 30 minutes recording data at 10 minute intervals to establish stability.

d. Reduce the saturated suction temperature to approximately 35 °F. Operate at these conditions until two consecutive sets of data indicate stable conditions, but in no case less than 15 minutes.
e. Increase the suction gas temperature entering the compressor by approximately 5 °F. Operate at these conditions until two consecutive sets of data indicate stable conditions, but in no case for less than 15 minutes.

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f. Repeat step e. until the compressor thermal protector actuates.

(1) Where a pilot type protector is employed, data shall be recorded immediately following actuation of the protector and this data shall be considered the protector actuation point. Then gradually return to the operating conditions of step c. Record one set of data when the protector resets. Operate the compressor at the initial conditions for a period of 30 minutes recording data at 10 minute intervals.

(2) Where an integral type protector is employed the last set of data taken before protector actuation shall be considered as the protector actuation point. When the protector resets the compressor shall be restarted and operated at the initial conditions of step c. for a minimum of 30 minutes, recording data at 10 minute intervals.

4.5.2.14.4 Failure criteria. One or more of the following shall constitute failure of this test:

a. Motor winding temperature in excess of manufacturer's recommended safe maximum temperature.
b. Compressor discharge temperature in excess of manufacturer's recommended safe maximum temperature.
c. Inability of compressor to operate satisfactorily throughout the test.

4.5.2.15 Voltage frequency variation. The compressor shall be installed in an air conditioner and the air conditioner placed in a test chamber capable of maintaining 125 °F +2 or -0 outdoor side and 90 °F +2 or -0 indoor conditions. The air conditioner shall be started and run for 1 hour for each of the following voltage and frequency conditions:

Test	Voltage	Frequency	Voltage				Frequency (Hz)	
a.	High	High	121	219	242	437	63	420
b.	Low	Low	109	197	218	395	47.5	380
c.	Low	High	109	197	218	395	63	420
d.	High	Low	121	219	242	437	47.5	380

Voltage tolerance ±1 percent.
Frequency tolerances ±1 percent.

Record data either continuously or at not greater than 15 minute intervals during each test. Data shall include:

a. Air temperatures: Dry bulb in and out of evaporator and condenser; wet bulb into the evaporator.
b. Refrigerant pressures: High and low sides of system.
c. Power characteristics: Voltage, current, watts.
d. Refrigerant conditions: Sight glass observation, temperature at entry to expansion device, at compressor discharge, at compressor suction, at thermal expansion valve bulb location.

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Failure of the compressor to start or operate at all the conditions shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a complete unit prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

124. Compressor not filled with oil and pressurized as specified for level A or commercial.
125. Lines not sealed as specified for level A or commercial.
126. Boxes not as specified for level A or B.
127. Compressor not secured within the box as specified for level A.
128. Compressors exceed the allowable weight for the box for level A or B.
129. Compressors of unlike description packed together for level A or B.
130. Strapping not as specified for level A or B.
131. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

5.2 Charging. Each compressor shall be filled to operating capacity with the oil recommended by the contractor and then pressurized with nitrogen to not less than 3 psig and not more than 15 psig holding charge and the lines sealed with sweat fittings to prevent leakage.

5.2.1 Suction, discharge and process lines. Suction and discharge lines shall be capped with copper and brazing in accordance with MIL-B-7883, grade B using brazing alloy conforming to QQ-B-654, BCup-5, grade IV or VI and flux

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conforming to O-F-499. Unless otherwise specified on drawings, the process tube shall be crimped and brazed by using the same brazing process, brazing alloy, and the flux used to braze suction and discharge lines.

5.3 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.3.1 Level A. Each compressor, charged and sealed as specified in 5.2, shall be placed in a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant, variety and grade as applicable to the weight of the compressor. The compressor shall be cushioned, blocked, and secured within the box in a manner to prevent movement and damage.

5.3.2 Commercial. Each compressor charged and sealed as specified in 5.2 shall be preserved in accordance with ASTM D 3951.

5.4 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.4.1 Level A. Compressors of like description, preserved as specified in 5.3, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class I, type I or IV, finish B, size as applicable.

5.4.2 Level B. Compressors of like description, preserved as specified in 5.3, shall be packed together as specified in 5.4.1, except boxes shall be domestic type or class as applicable. As an alternate, boxes conforming to PPP-B-640, class 2, style optional, or PPP-B-636, V3c, V11c, V13c, or V15c, style optional may be used when the weight of a single compressor or the combined weight of compressors to be packed together does not exceed the gross weight or size limitation of the box specification. Box closure and flat strapping reinforcement shall be in accordance with the appendix to the applicable box specification.

5.4.3 Commercial. Compressors, preserved a specified in 5.3, shall be packed in accordance with ASTM D 3951.

5.5 Marking.

5.5.1 Military. Marking for military levels of protection (level A or B) shall be marked in accordance with MIL-STD-129.

5.5.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container.

6. NOTES

6.1 Intended use. Compressors covered by this specification are intended for use in military air conditioners.

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TABLE V. Military air conditioners.

Compressor	Air Conditioner
Type I, Class I, Style I	MIL-A-52424
Type I, Class II, Style I	MIL-A-52424
Type I, Class I, Style II	MIL-A-52767
Type I, Class II, Style II	MIL-A-52767
Type II, Class I, Style II	MIL-A-52767
Type II, Class II, Style II	MIL-A-52767
Type II, Class III, Style II	MIL-A-52767
Type II, Class IV, Style II	MIL-A-52767
Type III, Class I, Style II	MIL-A-52767
Type III, Class II, Style II	MIL-A-52767
Type III, Class III, Style II	MIL-A-52767
Type IV, Class I, Style II	MIL-A-52435 MIL-A-52767
Type IV, Class II, Style II	MIL-A-52767
Type V, Class I, Style II	MIL-A-52767
Type V, Class II, Style II	MIL-A-52767

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style required (see 1.2).
c. Time frame required for submission of the first article pack. (see 5.1).
d. Levels of preservation and packing required (see 5.3 and 5.4).

6.3 Qualification.

6.3.1 End item procurement. When compressors covered by this specification are procured as end items by the Government, awards will be made only for compressors which, at the time set for the opening of bids or award of a negotiated contract are qualified under this specification and are qualified for inclusion in the Qualified Products List, regardless of whether the compressors have actually been so listed.

6.3.2 Component procurement. When the compressor covered by this specification is a component of an end item being procured by the Government, the requirements of FAR 1-1107.2(b) shall apply.

6.3.3 Qualifying activity. The attention of contractors is called to the above conditions and manufacturers are urged to arrange to have the compressors that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts or orders for compressors

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covered by this specification. The activity responsible for the Qualified Products List for this specification is: US Army Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FES, Fort Belvoir, VA 22060-5606.

6.3.4 Design data. Prior to the initiation of the qualification testing, the contractor shall furnish in writing the following information to the activity specified in 6.3.3.

a. Type, class, and style to be submitted (see 1.2).
b. Compressor manufacturer.
c. Serial numbers of compressors to be submitted for test.
d. Compressor manufacturer's model No.
e. Volts, frequency, phase.
f. Type oil.
g. Oil charge.
h. Motor winding resistance at _____°F.
i. Full load current.
j. Lock rotor current.
k. Maximum motor winding temperature.
l. Maximum refrigerant discharge temperature.
m. Performance curves indicating capacity, power consumption.
n. Motor manufacturer and model No.
o. Thermal protector cutout point.
p. Thermal protector manufacturer and part number.
q. Crankcase heater manufacturer and part number.

6.3.5 Qualification in air conditioners. Tests of the compressors when installed in air conditioners will be conducted by the U.S. Army Belvoir Research, Development and Engineering Center unless prior agreement is reached between the compressor contractor and the Government. In either case the compressor shall not be considered to be qualified until all tests in the applicable air conditioners are successfully completed.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the compressors in accordance with this specification.

6.5 Failure. For purposes of reliability testing a failure is defined as any malfunction of the compressor which causes it to perform outside of the requirements of this specification and is not caused by test peculiar conditions not expected to be seen in field use. The Government reserves the right to determine which incidents are considered failures. Malfunctions of thermal protection devices will be considered on an individual basis as to failure chargeability.

6.6 Subject term (key word) listing.

Air conditioner
Compressor
Compressor, air conditioner
Compressor, hermetic

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6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 99

Review activity:
Air Force - 82
DLA - GS

User activity:
Army -

Preparing activity:
Army - ME

Project 4130-0291

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52717D	Compressors, Hermetic, for Air Conditioner Application

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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